

Experiment - 3

Design a lexical Analyzer for given language should ignore the redundant spaces, tabs and new lines and ignore comments using C

CODE :

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
    char s[200], w[30];
    int i = 0, j;

    printf("Enter the string:\n");
    fgets(s, sizeof(s), stdin);

    while (s[i] != '\0')
    {
        if (isspace(s[i]))
            i++;

        else if (s[i] == '/' && s[i + 1] == '/')
            break;
```

```

else if (isalpha(s[i]))
{
    j = 0;
    while (isalnum(s[i]))
        w[j++] = s[i++];
    w[j] = '\0';

    if (!strcmp(w, "main") || !strcmp(w, "int") ||
        !strcmp(w, "char") || !strcmp(w, "float") ||
        !strcmp(w, "printf"))
        printf("%s is keyword\n", w);
    else
        printf("%s is identifier\n", w);
}

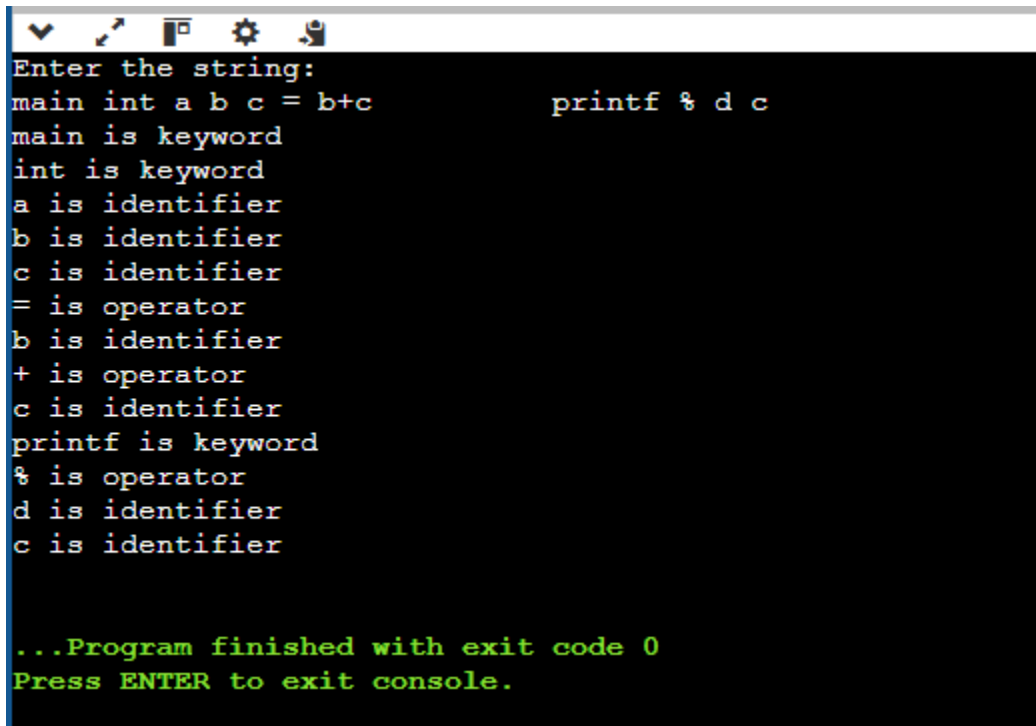
else if (s[i] == '=' || s[i] == '+' || s[i] == '-' ||
        s[i] == '*' || s[i] == '/' || s[i] == '%')
{
    printf("%c is operator\n", s[i++]);
}

else
    i++;
}

return 0;
}

```

OUTPUT :



```
Enter the string:
main int a b c = b+c           printf % d c
main is keyword
int is keyword
a is identifier
b is identifier
c is identifier
= is operator
b is identifier
+ is operator
c is identifier
printf is keyword
% is operator
d is identifier
c is identifier

...Program finished with exit code 0
Press ENTER to exit console.
```

Experiment - 4

Design a lexical Analyzer to validate operators to recognize the operators +,-,*,/ using regular arithmetic operators using C

CODE :

```
#include <stdio.h>
#include <string.h>
```

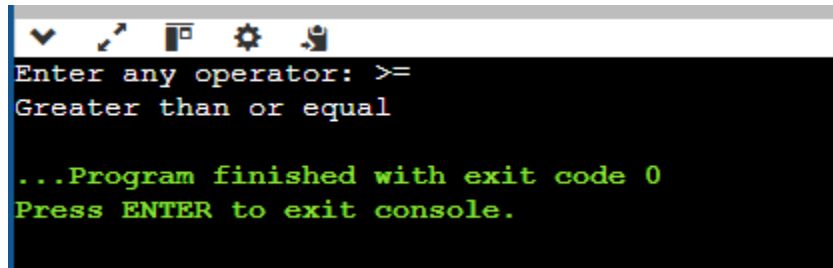
```
int main()
{
    char op[3];

    printf("Enter any operator: ");
    scanf("%s", op);

    if(!strcmp(op, "+"))
        printf("Addition");
    else if(!strcmp(op, "-"))
        printf("Subtraction");
    else if(!strcmp(op, "*"))
        printf("Multiplication");
    else if(!strcmp(op, "/"))
        printf("Division");
    else if(!strcmp(op, "<"))
        printf("Less than");
    else if(!strcmp(op, "<="))
        printf("Less than or equal");
    else if(!strcmp(op, ">"))
        printf("Greater than");
    else if(!strcmp(op, ">="))
        printf("Greater than or equal");
    else if(!strcmp(op, "=="))
        printf("Equal");
    else if(!strcmp(op, "!="))
        printf("Not equal");
    else
        printf("Invalid operator");
```

```
    return 0;  
}
```

OUTPUT :

A screenshot of a console window with a dark background and a light gray title bar. The title bar contains several icons: a downward arrow, a magnifying glass, a square, a gear, and a person. The console text is as follows:
Enter any operator: >=
Greater than or equal

...Program finished with exit code 0
Press ENTER to exit console.

```
Enter any operator: >=  
Greater than or equal  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```